

Byobu Terminal Multiplexer

Intelligent Robotics I

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Resources 2

What is Byobu? 4

Installation 9

Use 11

Resources

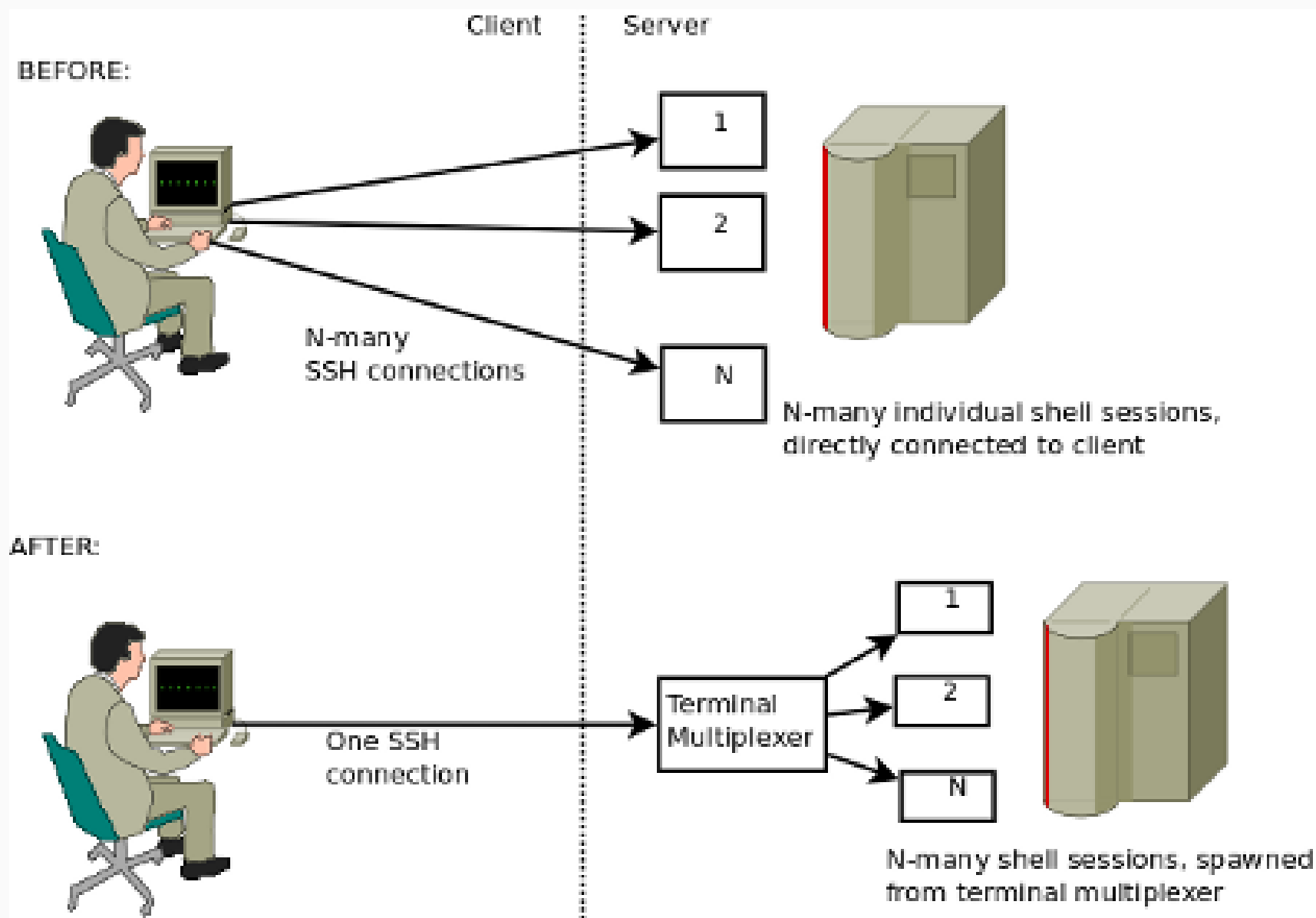
This presentation is based on the following resources:

- “Byobu Webpage” link

What is Byobu?



What is terminal multiplexing?



Source

What is Byobu?

```
def reload_required():
    try:
        if not os.path.exists(BYOBU_CONFIG_DIR):
            # 493 (decimal) is 0755 (octal)
            # Use decimal for portability across all python versions
            os.makedirs(BYOBU_CONFIG_DIR, 493)
        f = open(RELOAD_FLAG, 'w')
        f.close()
        if BYOBU_BACKEND == "screen":
            subprocess.call([BYOBU_BACKEND, "-X", "at", "0", "source", "%s/profile" % BYOBU_CONFIG_DIR])
    except:
        True

def terminal_size():
    # decide on some terminal size
    cr = ioctl_GWINSZ(0) or ioctl_GWINSZ(1) or ioctl_GWINSZ(2)
    # try open fds
    if not cr:
        # ...then ctty
        try:
            fd = os.open(os.ctermid(), os.O_RDONLY)
            cr = ioctl_GWINSZ(fd)
            os.close(fd)
        except:
            pass
    if not cr:
        # env vars or finally defaults
        try:
            cr = (env['LINES'], env['COLUMNS'])
        except:
            cr = (25, 80)
    # reverse rows, cols
    return int(cr[1] - 5), int(cr[0] - 5)

def menu(snackScreen, size, isInstalled):
    if isInstalled:
        installtext = _("Byobu currently launches at login (toggle off)")
    else:
        installtext = _("Byobu currently does not launch at login (toggle on)")
    84,0-1 23%
```

```
Source: byobu
Section: misc
Priority: optional
Maintainer: Dustin Kirkland <kirkland@ubuntu.com>
Uploaders: Antoine Beaupré <anarcat@koumbit.org>
Standards-Version: 3.9.5
Build-Depends: debhelper (>= 7.0.50~), gettext-base, automake, autoconf, pep8
Homepage: http://byobu.co
Vcs-Bzr: http://bazaar.launchpad.net/~kirkland/byobu/trunk
```

```
Package: byobu
Architecture: all
Depends:
    ${misc:Depends}, ${perl:Depends}, ${python:Depends},
    debconf (>= 0.5) | debconf-2.0,
    gettext-base,
    python3|python,
    python3-newt|python-newt,
    tmux (>= 1.5) | screen,
    gawk
Recommends:
```

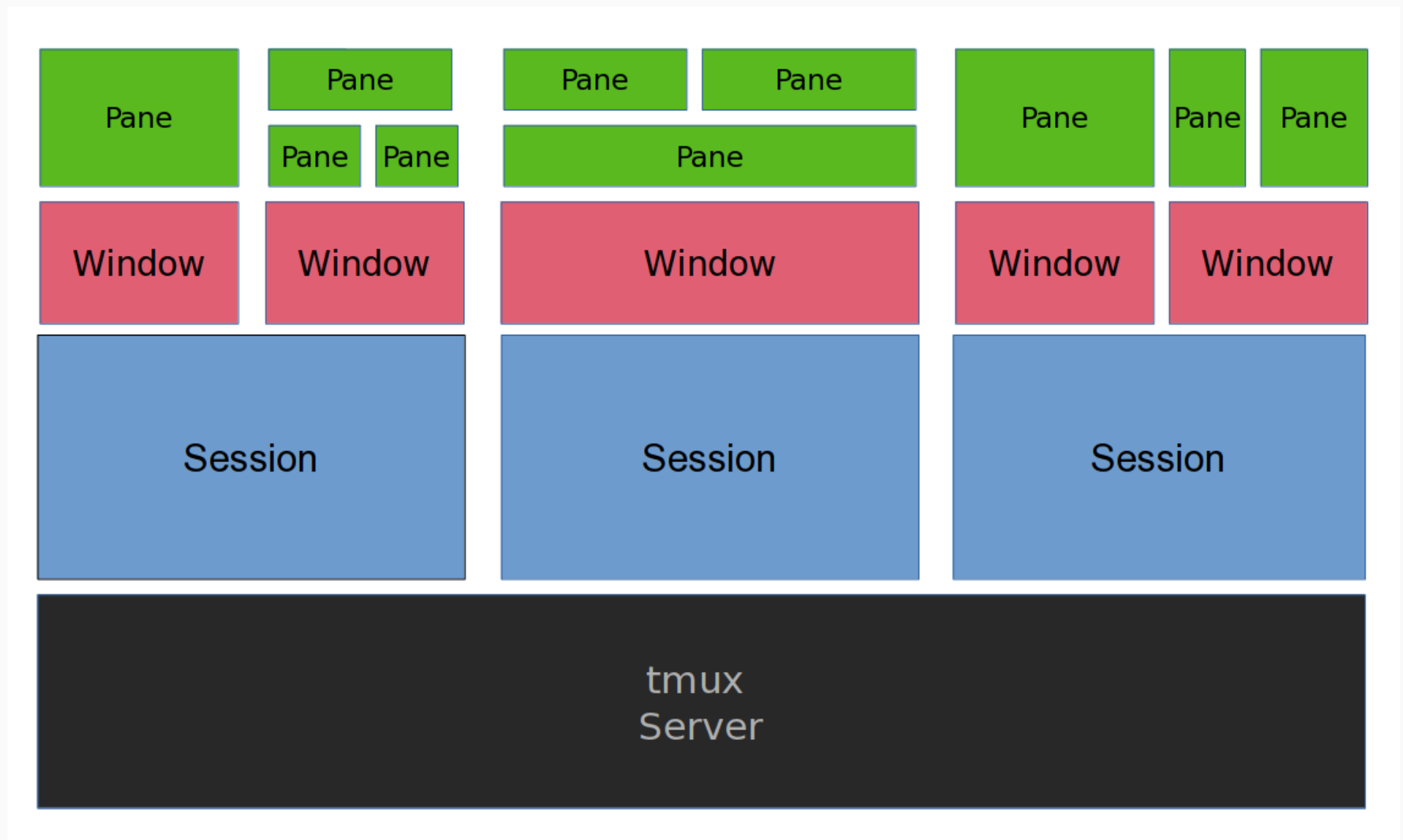
1,3 Top

```
top - 15:39:14 up 3:52, 2 users, load average: 0.39, 0.45, 0.54
Tasks: 277 total, 1 running, 276 sleeping, 0 stopped, 0 zombie
%Cpu(s): 7.6 us, 4.1 sy, 0.0 ni, 88.3 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem: 16122772 total, 5656816 used, 10465956 free, 114128 buffers
KiB Swap: 0 total, 0 used, 0 free. 1708780 cached Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2709	kirkland	9	-11	439872	7976	5288	S	10.3	0.0	5:01.30	pulseaudio
3362	kirkland	20	0	3226056	228996	76136	S	4.6	1.4	9:48.11	chromium-browser
3875	kirkland	20	0	1789036	44348	12276	S	3.3	0.3	1:54.39	chromium-browser
2812	kirkland	20	0	1378972	73732	31888	S	2.7	0.5	3:53.17	compiz
3422	kirkland	20	0	1882896	284188	31060	S	2.3	1.8	1:49.46	chromium-browser
5150	kirkland	20	0	1682228	284732	34576	S	2.0	1.8	0:13.76	chromium-browser
1466	root	20	0	319616	41428	31728	S	1.3	0.3	2:51.85	Xorg
4039	kirkland	20	0	877424	31176	15108	S	1.3	0.2	2:00.32	gnome-terminal
4076	kirkland	20	0	35788	13240	1296	S	1.3	0.1	1:15.10	tmux
6033	kirkland	20	0	1518840	110824	27448	S	1.3	0.7	2:08.29	chromium-browser
2552	kirkland	20	0	437896	7108	2932	S	1.0	0.0	1:13.68	ibus-daemon
2534	kirkland	20	0	41476	2916	916	S	0.7	0.0	0:12.44	dbus-daemon
2657	kirkland	20	0	719796	23784	12196	S	0.7	0.1	0:26.65	unity-panel-ser
3709	kirkland	20	0	1153792	41084	31252	S	0.7	0.3	0:12.94	chromium-browser
6522	kirkland	20	0	29152	1764	1176	R	0.7	0.0	0:00.37	top

Source

Under the Hood



Source

- multiple windows and panes
- persistence (detach + reboot + reattach)
- session sharing

Installation



Ubuntu

```
sudo apt install byobu
```

MacOS

```
brew install byobu
```

Launch

```
byobu
```

Use



Shift-F1 show commands sorted by key

```
byobu is a suite of enhancements to tmux, as a command line
tool providing live system status, dynamic window management,
and some convenient keybindings:

F1                               * Used by X11 *
  Shift-F1                       Display this help
F2                               Create a new window
  Shift-F2                       Create a horizontal split
  Ctrl-F2                        Create a vertical split
  Ctrl-Shift-F2                  Create a new session
F3/F4                           Move focus among windows
  Alt-Left/Right                Move focus among windows
  Alt-Up/Down                   Move focus among sessions
  Shift-Left/Right/Up/Down      Move focus among splits
  Shift-F3/F4                   Move focus among splits
  Ctrl-F3/F4                    Move a split
  Ctrl-Shift-F3/F4              Move a window
  Shift-Alt-Left/Right/Up/Down  Resize a split
F5                               Reload profile, refresh status
  Alt-F5                        Toggle UTF-8 support, refresh status
  Shift-F5                      Toggle through status lines
  Ctrl-F5                       Reconnect ssh/gpg/dbus sockets
  Ctrl-Shift-F5                 Change status bar's color randomly
F6                               Detach session and then logout
  Shift-F6                      Detach session and do not logout
  Alt-F6                        Detach all clients but yourself
  Ctrl-F6                       Kill split in focus
F7                               Enter scrollbar history
  Alt-PageUp/PageDown           Enter and move through scrollbar
  Shift-F7                      Save history to $BYOBU_RUN_DIR/printscr
on
F8                               Rename the current window
  Ctrl-F8                      Rename the current session
  Shift-F8                     Toggle through split arrangements
  Alt-Shift-F8                 Restore a split-pane layout
  Ctrl-Shift-F8                Save the current split-pane layout
F9                               Launch byobu-config window
```

A pane is a terminal.

key	command
Shift-F2	split window vertical
Shift-F3/F4	move focus panes
Shift-Alt-arrow	resize split
Ctrl-F2	split windows horizontal
Ctrl-F3/F4	move split
Ctrl-F6	kill pane in focus
Shift-F11	zoom into/out split
Alt-F11	expand to full window

A window is a set of panes.

key	command
F2	new window
F3/F4	move focus window
Ctrl-Shift-F3/F4	move window
Alt-Left/Right	move focus window
F8	rename current window

A session is a set of windows.

key	command
Shift-Ctr-F2	create new session
Alt-Up/Down	move focus between sessions
Ctr-F8	rename session

The status bar is the bar at the bottom.

key	command
F5	refresh
Shift-F5	cycle status lines
Ctrl-Shift-F5	change color
F9	configure status bar

key	command
F6	detach and logout
Ctrl-F6	kill all panes to kill session

key	command
Alt-PageUp	enter scrollback
/	enter vi search inside scrollback
Shift-F7	save scrollback to file

The mouse can be used inside Byobu terminals.

key	command
Alt-F12	toggle on/off mouse support
Shift	support to select text without mouse